

Laboratory Guide 7: Quantum Optics — Photons, Interference, and Entanglement

The Hardware of Quantum Technology

Course: Quantum Computing Foundation

1 Introduction and Objectives

Labs 5 and 6 gave you the mathematics of qubits and a protocol built on it. This lab shows you the **hardware those abstractions run on**. A single photon *is* a qubit — twice over: its **path** through an interferometer is one two-level system, its **polarization** is another. Every device on the optical bench — beam splitter, phase shifter, polarizer — is one of the operators you already know, executed by glass and crystal. And every strange number this lab produces (a coincidence counter pinned at zero, two 50/50 elements composing to certainty, a filter that *revives* blocked light, twin photons that always agree) is a Born-rule computation you have already done abstractly.

Prerequisites:

- *Fundamental Concepts of Quantum Technologies* (lecture series).
- Lab 5: *Quantum Measurements* — the Born rule, measurement Procedures 1 and 2, and the states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$ on the Bloch sphere.
- Lab 6: *Quantum Key Distribution (BB84)* — mutual unbiasedness and the measurement–disturbance–security bridge.
- Familiarity with Dirac notation: $|\psi\rangle, \langle\phi|, \langle\phi|\psi\rangle$.

Connections to Previous Labs

The Lab 5–6 bridge. The path states $|A\rangle, |B\rangle$ and the polarization states $|H\rangle, |V\rangle$ are both the Lab 5 computational (Z) basis; the diagonal polarizations at $\pm 45^\circ$ are exactly the Lab 5 X -basis states $|+\rangle, |-\rangle$. The four BB84 signal states of Lab 6 are *physically* $|H\rangle, |V\rangle, |+\rangle, |-\rangle$ — Part C of this lab is the transmitter and receiver hardware of a real QKD link, and Part B’s which-path result is Lab 6’s “Eve leaves fingerprints” theorem restated as an identity about interference.

Mode: Interactive Optical Bench + Qiskit **Credits:** 1 ECTS **Estimated time:** ≈ 10 h

Learning Objectives:

1. **Explain the operational principles of single-photon and entangled-photon experiments** — beam splitters as unitaries, detection as projective measurement, anti-correlation, interference, and coincidence counting.
2. **Design and analyze simple optical setups** — predict the output statistics of a Mach–Zehnder interferometer, a polarizer cascade, or a two-analyzer coincidence experiment, and verify the prediction photon by photon in the simulator.

3. **Interpret interference and coincidence data** — fringe visibility, which-path complementarity ($\mathcal{V} = s$), and the quantitative gap between quantum and classical-wave predictions.
4. **Discuss the role of optical platforms in quantum communication and computation** — polarization-encoded and entanglement-based QKD, interferometric sensing, and dual-rail photonic quantum computing.

Simulator

The lab’s interactive page ([lab7_optics.html](#)) contains the **Optical Bench** built for this lab: five guided presets reproduce exactly the experiments of this guide — **Preset A** (beam splitter + coincidence counter), **Preset B1** (Mach–Zehnder fringes), **Preset B2** (which-path marker), **Preset C** (polarizer cascade), **Preset D** (entangled pairs, optional). Every displayed probability comes from the same Born-rule pipeline as the formulas in this guide, and all runs are seeded: the same seed and the same sequence of actions always reproduce the same counts.

This guide is written simulator-first. Each Part opens with an “Open the Simulator First” box. Do that two-minute run *before* reading the theory: the counts you observe pose exactly the question the section then answers. The companion Qiskit notebook rebuilds the same optics as quantum circuits:

https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab7/lab7_optics.ipynb

2 Part A: One Photon at a Beam Splitter

Open the Simulator First

Open the bench in **Preset A**: a single-photon source, one 50/50 beam splitter, a detector on each output port, and a coincidence counter wired to both. Fire **10 000 photons** and watch two things: the two detector tallies, and the coincidence counter. The tallies drift toward 50/50 — but the coincidence counter *never moves*. Keep that second observation in mind; it is the whole section.

The Question

What does it actually mean that light comes in photons — and what happens when one indivisible photon hits a half-silvered mirror?

Counterintuitive — and Real

One photon **takes both paths** — the state after the beam splitter is a superposition over the two output arms, and Part B will prove it by interference — yet it **never triggers both detectors**. Not rarely: *never*. The photon is divisible in amplitude but indivisible in detection.

A half-silvered mirror (a **50/50 beam splitter**) transmits half the light and reflects half. For a classical wave that sentence is unambiguous: the energy splits, half into each arm. But light arrives in indivisible quanta. A single photon cannot deliver half a click to each detector — so what does “splits 50/50” mean for *one photon*?

Quantum mechanics answers with a state, not a story. Write the two output paths as the basis states $|A\rangle$ (transmitted) and $|B\rangle$ (reflected) — the same two-level system as any Lab 5 qubit. A lossless beam splitter must act as a **unitary** on this qubit (unitarity here *is* energy conservation — the photon must exit somewhere), and the symmetric 50/50 choice is:

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad \Rightarrow \quad B|A\rangle = \frac{|A\rangle + i|B\rangle}{\sqrt{2}}. \quad (1)$$

Theory Recap

Why the i ? A naive “50/50 without phase”, $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, is a singular matrix — it is *not a possible physical device*, because it would merge two distinguishable inputs into one output and lose a photon in the process. The phase i on reflection is the minimal repair that restores unitarity (Zeilinger, 1981). This phase convention — i on reflection, used consistently — is fixed for the entire lab: every formula below is derived under it, and the simulator uses exactly this matrix.

Detection at the two output ports is a path-basis measurement — **Procedure 1** from Lab 5, executed destructively (the photon is absorbed by the detector that clicks). The Born rule gives the two outcome probabilities:

$$P(\text{detector at port A}) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad P(\text{detector at port B}) = \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad (2)$$

$$P(\text{both detectors}) = 0 \quad \text{exactly.} \quad (3)$$

That last line deserves a hard look. The state $(|A\rangle + i|B\rangle)/\sqrt{2}$ is **one photon in superposition over two paths** — **not two photons**. A path measurement has exactly two mutually

exclusive outcomes; “both detectors click” is not a low-probability event, it is *not in the outcome set at all*. The zero is structural, not statistical.

Formula → Number → Bench

$P = \frac{1}{2}, \frac{1}{2}$ and $P_{\text{coinc}} = 0$. In **Preset A**: after 10 000 photons the tallies land within statistical noise of **5 000 / 5 000** (the theory overlay shows the exact 50% line), while the coincidence counter reads **0** — **exactly, after every single photon**. Fire another 10 000 and check that the zero is not luck.

Important

The honest classical contrast. A classical wave or pulse really does split: half the energy in each arm, and with two detectors both can fire in the same trial. The standard figure of merit is $g^{(2)}(0)$, the normalized zero-delay second-order correlation — the coincidence rate normalized to random chance (Loudon, 2000). Semiclassical models predict coincidence rates at least at the random-chance level, $g^{(2)}(0) \geq 1$, while the one-photon state gives $g^{(2)}(0) = 0$. Grangier, Roger and Aspect measured exactly this anti-correlation in 1986; it is the operational meaning of “the photon is indivisible”, and it is the first of this lab’s four experimental pillars.

Real-World Anchor

Where this is an industry, not a thought experiment. One photon distributed over two spatial modes is the **dual-rail qubit** — the native encoding of photonic quantum computing. **Quandela** builds the single-photon sources (the “fire one photon on demand” box the simulator idealizes) and dual-rail photonic processors around exactly this $\{|A\rangle, |B\rangle\}$ qubit; **ORCA Computing** builds photonic quantum processors on the same principle that beam-splitter networks perform unitary computation on light. The linear-optical quantum computing literature (the KLM protocol) showed that such networks plus detection are universal. When you fire photons at the bench, you are operating the elementary cell of those machines.

3 Part B: The Mach–Zehnder Interferometer and Which-Path Information

Open the Simulator First

Switch the bench to **Preset B1**: two beam splitters back to back, mirrors folding the two arms together, and a phase knob φ on arm B. Leave $\varphi = 0^\circ$ and fire 10 000 photons. Every photon exits at **D1** — 100/0, from a device built of two 50/50 elements. Now sweep the φ slider and watch the split move. The counts *depend on the phase difference between the two paths*.

The Question

If the photon went one way or the other, the detector counts could not depend on the path it didn’t take. They do. What now?

Counterintuitive — and Real

Two 50/50 beam splitters in sequence give a **100/0** outcome. Model each splitter as an independent coin flip — the reasonable move if the photon “really” takes one path — and you predict 50/50 at the output. The observed perfect nulls and perfect peaks are the single-photon interference signature: the amplitudes of *both* paths add before anything is detected.

The **Mach–Zehnder interferometer** is the workhorse: beam splitter BS1 opens two paths, mirrors bring them back together, a **phase shifter** $\Phi(\varphi) = \text{diag}(1, e^{i\varphi})$ on arm B models any path-length difference, and beam splitter BS2 recombines. (The mirrors add the same phase to both arms — a global phase, dropped.) Step the input $|A\rangle$ through the device:

$$|A\rangle \xrightarrow{\text{BS1}} \frac{|A\rangle + i|B\rangle}{\sqrt{2}} \xrightarrow{\Phi(\varphi)} \frac{|A\rangle + ie^{i\varphi}|B\rangle}{\sqrt{2}} \xrightarrow{\text{BS2}} \frac{(1 - e^{i\varphi})|A\rangle + i(1 + e^{i\varphi})|B\rangle}{2}. \quad (4)$$

With detector **D1** on output port B and **D2** on output port A, the Born rule and the identity $|1 \pm e^{i\varphi}|^2 = 2 \pm 2\cos\varphi$ give the fringes:

$$P(D1) = \cos^2\frac{\varphi}{2}, \quad P(D2) = \sin^2\frac{\varphi}{2}. \quad (5)$$

At $\varphi = 0$: $P(D1) = 1$ — the 100/0 you just observed. Algebraically this is the composition identity $B^2 = iX$: two symmetric beam splitters are a perfect path swap (up to an overall phase). At $\varphi = \pi$ the same instrument, retuned, sends every photon to D2 instead. Between the extremes the split interpolates as a perfect cosine:

phase φ	0°	60°	90°	180°
$P(D1) = \cos^2(\varphi/2)$	1	3/4	1/2	0
$P(D2) = \sin^2(\varphi/2)$	0	1/4	1/2	1

Table 1: Mach–Zehnder anchor points, equation (5). The 60° column is the 3:1 fringe point used by Preset B1.

For fringes of the form $P(\varphi) = (1 + \mathcal{V}\cos\varphi)/2$, the **fringe visibility** is

$$\mathcal{V} = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}}, \quad (6)$$

and the unmarked interferometer has $P_{\max} = 1$, $P_{\min} = 0$: $\mathcal{V} = 1$.

Formula → Number → Bench

At $\varphi = \pi/3$ (60°): $P(D1) = \cos^2(30^\circ) = \frac{3}{4}$ — a **75% / 25%** split. In **Preset B1**, set the slider to 60° and fire 10 000 photons: the tallies land on 75/25 within noise, on top of the theory overlay. Then return to $\varphi = 0^\circ$ and re-verify the **100/0** — two 50/50 splitters, certain outcome.

Real-World Anchor

The fringe as a measuring instrument. φ is optical path-length difference in disguise. Any physical quantity that stretches one arm — rotation in a **fiber-optic gyroscope** (the navigation-grade rotation sensors in aircraft), strain, temperature, or a passing gravitational wave in the kilometer-scale **LIGO-class interferometers** — shifts the fringe. Near the steepest point of the cosine, reading $P(D1)$ to precision δP reads the phase to $\delta\varphi \approx 2\delta P$: counting photons *is* the measurement. That is interferometric sensing, and it is this exact circuit.

3.1 Which-Path Marking: Complementarity, Quantified

Open the Simulator First

Switch to **Preset B2**: the same interferometer, plus a **which-path marker** in the arms — a probe qubit that records which arm the photon took. Its dial is the **marker overlap** s : at $s = 1$ the marker states for the two arms are identical (no record); at $s = 0$ they are orthogonal (a perfect record). Set $s = 0$, sweep φ , and watch the fringes: they are **gone** — flat 50/50 at every phase. Now slide s upward and watch them come back, exactly in proportion.

Counterintuitive — and Real

An **unread** which-path marker still destroys the fringes. Nobody looks at the record. It is enough that the record *exists* — that the universe could in principle tell which path was taken. Interference is lost the moment the paths become distinguishable, not the moment someone learns the answer.

Model the marking honestly. Between the beam splitters, couple the path qubit to a marker qubit: path A tags the marker into state $|M_A\rangle$, path B into $|M_B\rangle$ (normalized, with overlap $s = \langle M_A|M_B\rangle$ taken real in $[0,1]$). We write the two-qubit state as $|\text{path marker}\rangle$ with the **path qubit leftmost** — big-endian ordering, qubit 0 is the leftmost symbol; the companion notebook converts Qiskit’s opposite bitstring order before displaying anything. After BS1 and the marking:

$$\frac{|A\rangle|M_A\rangle + ie^{i\varphi}|B\rangle|M_B\rangle}{\sqrt{2}}. \quad (7)$$

Recombine at BS2 (acting on the path qubit only) and compute $P(D1)$, summing over the marker — read or unread, the arithmetic cannot tell:

$$P(D1; \varphi, s) = \frac{1}{4} \left\| |M_A\rangle + e^{i\varphi}|M_B\rangle \right\|^2 = \frac{1 + s \cos \varphi}{2} \implies \boxed{\mathcal{V} = s = |\langle M_A|M_B\rangle|} \quad (8)$$

Fringe visibility equals the indistinguishability of the marker states. $s = 1$: no record, full fringes. $s = 0$: perfect record, no fringes — $P(D1) = \frac{1}{2}$ at every phase, as if the photon really had flipped a coin at BS1. In between, the trade is exactly linear in the fringe term. A perfect marking ($s = 0$) is physically a controlled-NOT — the path controls the marker, turning the superposition into the entangled path-marker state $(|00\rangle + ie^{i\varphi}|11\rangle)/\sqrt{2}$ (path leftmost, as always) — and reading the marker afterwards is a genuine path measurement (**Procedure 1** on the marker qubit): same statistics as putting a detector in the arm, but the photon survives to reach BS2.

Formula $\rightarrow$ Number $\rightarrow$ Bench

At $s = 0.5$: $\mathcal{V} = 0.5$, so the fringes run between $P_{\max} = \frac{1+0.5}{2} = \mathbf{0.75}$ and $P_{\min} = \frac{1-0.5}{2} = \mathbf{0.25}$. In **Preset B2**, set $s = 0.5$, fire at $\varphi = 0^\circ$ ($\rightarrow 75\%$) and at $\varphi = 180^\circ$ ($\rightarrow 25\%$), and read the visibility card: it converges to 0.5 — the dial you set, recovered from counting statistics alone.

Connections to Previous Labs

Lab 5 said “measurement disturbs the state”; Lab 6 said “Eve’s measurement leaves fingerprints”. Both are this one identity $\mathcal{V} = |\langle M_A | M_B \rangle|$: an eavesdropper who gains path information ($s < 1$) *necessarily* degrades the interference any honest party could observe. Information out $\implies$ coherence lost, in exact proportion. (The general form is Englert’s duality relation $\mathcal{V}^2 + \mathcal{D}^2 \leq 1$ with distinguishability $\mathcal{D} = \sqrt{1 - s^2}$ — our ideal marker saturates it.)

4 Part C: Polarization — Quantum Malus and the Three-Polarizer Surprise

Open the Simulator First

Switch to **Preset C**: an unpolarized source and a rack of polarizing filters. Start with two **crossed** polarizers — 0° and 90° . Fire 10 000 photons: **nothing gets through**. Now insert a 45° polarizer between them and fire again: **12.5%** of the photons pass all three filters. You added an absorber, and the light came back.

The Question

Two crossed filters block everything. Insert a third filter between them and light comes through. Why does adding an absorber *increase* transmission?

Counterintuitive — and Real

Adding an absorbing polarizer raises transmission from exactly **0** to **1/8**. No sieve, filter, or “the polarizer lets through the right components” picture survives this: a filter that only ever *removes* light cannot make more light arrive. Something other than removal is happening. (You can try this at home with three lenses from polarized sunglasses.)

Polarization is the photon’s second qubit: $|H\rangle$ (horizontal) and $|V\rangle$ (vertical) form the computational basis, and the state at linear angle θ from the H-axis is $|\theta\rangle = \cos\theta |H\rangle + \sin\theta |V\rangle$. The diagonal states at $\pm 45^\circ$ are exactly the Lab 5 X -basis pair $|\pm\rangle$.

An ideal polarizer at angle θ is the projector $\Pi_\theta = |\theta\rangle\langle\theta|$, and this is the crucial fact: a polarizer is an inherently **Procedure-1 device**. Each photon either passes — *and is then in state* $|\theta\rangle$ — or is absorbed. There is no fractional photon. For a photon in $|H\rangle$ meeting a polarizer at θ , the Born rule gives the **quantum Malus law**:

$$P(\text{pass}) = |\langle\theta|H\rangle|^2 = \cos^2\theta, \quad \text{post-state } |\theta\rangle. \quad (9)$$

polarizer angle θ	0°	30°	45°	60°	90°
$P(\text{pass})$ from $ H\rangle$	1	$3/4$	$1/2$	$1/4$	0

Table 2: Quantum Malus law, equation (9), at the five anchor angles.

Classical Malus ($I = I_0 \cos^2 \theta$) reads identically — but the quantum version is **per-photon and probabilistic**: $\cos^2 \theta$ is a Born probability, the same $|\langle m|\psi\rangle|^2$ as every measurement since Lab 5, and each individual photon entirely passes or is entirely absorbed.

Formula → Number → Bench

$\cos^2(60^\circ) = 25\%$. In **Preset C**, switch the source to $|H\rangle$, enable a single polarizer at 60° , and fire 10 000 photons: the pass fraction lands on 25% within noise. Then rebuild the crossed pair (unpolarized source, 0° and 90°) → **exactly 0** — and insert the 45° filter → $1/8 = 12.5\%$.

Now the resolution of the three-polarizer surprise. For a cascade $\theta_1, \dots, \theta_N$, each passage *renews* the state to the current polarizer’s $|\theta_j\rangle$ — the Procedure-1 post-state. So the pass probabilities multiply with the *differences* between consecutive angles:

$$P(\text{pass all}) = P_1 \cdot \prod_{j=2}^N \cos^2(\theta_j - \theta_{j-1}), \quad (10)$$

with $P_1 = \frac{1}{2}$ for an unpolarized source (no polarization preference — every first filter passes half). The two experiments you ran:

$$\text{crossed } (0^\circ, 90^\circ) : \frac{1}{2} \cdot \cos^2(90^\circ) = 0, \quad \text{three } (0^\circ, 45^\circ, 90^\circ) : \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}. \quad (11)$$

The middle filter does not “let more light through” — it **rewrites the state**. After the 45° polarizer the photon is in $|+\rangle$, which is no longer orthogonal to the final 90° filter. The projective post-state — the fact that measurement *changes* the system, the disturbance you met in Lab 5 — is the entire explanation. The 45° polarizer is an X -basis measurement of the polarization qubit, executed by a plastic film.

Procedure 1 vs Procedure 2

Two physically different devices. Lab 5 distinguished two ways to measure in the diagonal basis, and in optics they are *different hardware*:

- **Procedure 1** — a polarizer at 45° : probabilities $|\langle \pm|\psi\rangle|^2$, passed photon left in $|+\rangle$ (the $|-\rangle$ outcome is an absorbed photon).
- **Procedure 2** — a half-waveplate at 22.5° (which implements the canonical basis-change rotation $U_X = H$) followed by a polarizing beam splitter and two detectors: *identical probabilities*, but the photon exits in $|H\rangle$ or $|V\rangle$, not $|\pm\rangle$.

Same statistics, different post-measurement states: the overlap between the two devices’ “+” outputs is $|\langle +|H\rangle|^2 = \frac{1}{2}$, not 1. Downstream optics see different photons. Lab 6’s naive-resend attack — where Eve’s forgotten re-encoding turned a 25% error rate into 37.5% — was exactly this distinction carrying a security consequence.

Real-World Anchor

This hardware runs QKD networks today. Polarization-encoded BB84 uses precisely these components: Lab 6’s four signal states *are* $|H\rangle, |V\rangle, |+\rangle, |-\rangle$; Alice’s transmitter is a polarization preparer, Bob’s receiver is a basis-switched analyzer — a rotation plus a polarizing beam splitter, Procedure 2 in glass. Metropolitan QKD fiber networks (deployed in Europe and Asia, and the building block of the planned **EuroQCI** infrastructure) run this Part’s optics at telecom scale: every “secure key bit” started life as a photon meeting a polarization analyzer.

Deep Dive

The polarizer staircase. Generalize the insertion trick: between 0° and 90° , insert intermediate filters in equal steps of $90^\circ/N$ — N steps, hence $N + 1$ polarizers counting both ends. The transmission $\frac{1}{2} \cos^{2N}(90^\circ/N)$ *rises* toward $\frac{1}{2}$ as N grows: many gentle projections approximate a lossless rotation of the polarization. ($N = 2$ is the three-polarizer experiment.) Try the staircase $0^\circ, 30^\circ, 60^\circ, 90^\circ$ in Preset C: $\frac{1}{2} (3/4)^3 = 27/128 \approx 21.1\%$, already far above the 12.5% of the single insertion.

5 Part D: Entangled Photon Pairs (Optional)

This part is optional enrichment: nothing in the activities’ core depends on it. It is also the doorway to the deepest experimental result in the field — read on if you want to see it.

Open the Simulator First

Switch to **Preset D**: a crystal that emits **pairs** of photons, one to each side, each side ending in an analyzer (angle dial + two detectors, “+” and “−”). Set both dials to the **same angle** — any angle — and fire 10 000 pairs. Each side alone clicks “+” or “−” like a fair coin. But the **agreement** counter reads **100%**: the two sides always give the same answer. Now change the common angle and fire again — still 100%.

The Question

Two photons, each individually a perfect coin toss, always agree when asked the same question — at any angle, chosen after they separated. What kind of correlation is that?

Counterintuitive — and Real

Entangled photons are **individually random at every angle, perfectly correlated whenever the angles match** — and the best classical wave model is not just vaguely wrong about this, it is off by **exactly a factor of 2** in the correlation amplitude. The simulator plots both curves; the gap between them is the measurement.

Nonlinear crystals (spontaneous parametric down-conversion, the workhorse entangled-pair source since the 1990s — Kwiat et al., 1995) emit two photons in the joint polarization state

$$|\Phi^+\rangle = \frac{|HH\rangle + |VV\rangle}{\sqrt{2}}, \quad (12)$$

written photon 1 leftmost, photon 2 rightmost — the same leftmost-first, big-endian ordering as Part B’s $|\text{path marker}\rangle$ kets. Each side passes a two-channel analyzer at angle α (photon 1)

or β (photon 2): outcome “+” means the photon exited the $|\theta_{\text{analyzer}}\rangle$ port, “−” the orthogonal port. Two facts, both exact:

- **Each photon alone: a perfect coin.** Summing over the partner’s outcomes leaves photon 1 (or 2) in the maximally mixed state $I/2$ — *exactly* unpolarized. $P(+)=\frac{1}{2}$ at *every* analyzer angle. No analyzer setting on one side is visible in the other side’s single rates.
- **The pair together: locked.** The joint amplitude is $\langle\alpha, \beta|\Phi^+\rangle = (\cos\alpha\cos\beta + \sin\alpha\sin\beta)/\sqrt{2} = \cos(\alpha - \beta)/\sqrt{2}$, so with $\Delta = \alpha - \beta$:

$$P(++)=P(--)=\frac{\cos^2\Delta}{2}, \quad P(+-)=P(-+)=\frac{\sin^2\Delta}{2}. \quad (13)$$

At matched angles ($\Delta = 0$): agreement with certainty. At $\alpha = 0^\circ$, $\beta = 22.5^\circ$, for instance: $P(++)=P(--)=(1+\cos 45^\circ)/4 \approx 0.4268$ and $P(+-)=P(-+)=(1-\cos 45^\circ)/4 \approx 0.0732$ — the four sum to 1.

The standard summary is the **correlation function** $E(\alpha, \beta) = P(++)+P(--)-P(+-)-P(-+)$, which collapses to a function of the angle *difference* alone — the state $|\Phi^+\rangle$ is invariant when both analyzers rotate together:

$$E(\alpha, \beta) = \cos 2\Delta, \quad \Delta = \alpha - \beta. \quad (14)$$

analyzer difference Δ	0°	22.5°	45°	90°
$E = \cos 2\Delta$	1	$\sqrt{2}/2 \approx 0.707$	0	−1

Table 3: Correlation anchors, equation (14): perfect correlation, the $\sqrt{2}/2$ point, mutually unbiased analyzers, perfect anti-correlation.

Formula → Number → Bench

$E(0^\circ, 22.5^\circ) = \cos 45^\circ = \frac{\sqrt{2}}{2} \approx \mathbf{0.707}$. In **Preset D**, set $\alpha = 0^\circ$, $\beta = 22.5^\circ$, fire 10 000 pairs, and read the measured E against the theory overlay. Then set the dials equal ($\Delta = 0$) and confirm **100% agreement** — against the classical-wave overlay’s **75%** — while each side’s own counter stays at 50/50. At $\Delta = 45^\circ$, $E = 0$: the two analyzers have become mutually unbiased — Lab 6’s key concept reappearing as zero correlation.

Why is this *deep* and not just pretty? Try the obvious classical story: each pair is emitted with a shared random polarization λ , and each analyzer responds by classical Malus, $\cos^2$ of the angle to λ . Average over uniform λ : expanding each factor via $\cos^2 x = (1 + \cos 2x)/2$, the single-cosine averages vanish, and the lone surviving average is $\langle \cos 2(\alpha - \lambda) \cos 2(\beta - \lambda) \rangle_\lambda = \cos(2\Delta)/2$, so

$$P_{\text{cl}}(++) = \frac{1}{4} + \frac{1}{8} \cos 2\Delta \quad \implies \quad E_{\text{cl}}(\Delta) = \frac{\cos 2\Delta}{2} \quad (15)$$

— the same shape, but **half the size**. At matched analyzers the classical wave model manages only 75% agreement ($E_{\text{cl}}(0) = \frac{1}{2}$) against the observed 100%. This is not a philosophical complaint — it is a **device-verifiable factor of 2**, and the simulator’s overlay draws both curves so you can watch the measured points choose the quantum one. More sophisticated classical models (predetermined answers for every angle) can patch the matched-angle agreement but then fail elsewhere on the $\cos 2\Delta$ curve — that is **Bell’s theorem** (Bell, 1964; Clauser–Horne–Shimony–Holt, 1969), which we cite rather than derive: no local classical story reproduces the

full curve. What this lab establishes at the correlation level is the honest core: the measured curve is $\cos 2\Delta$, and the classical wave prediction misses it by a factor of 2.

Real-World Anchor

Entanglement as infrastructure. The pair source you are firing is the physical basis of **entanglement-based QKD** (the BBM92 protocol): Alice and Bob each receive one photon of a pair, measure in randomly switched bases, and the matched-angle perfect correlations play the role that Alice's prepared states played in Lab 6's BB84 — with the added property that the correlations themselves certify that no third party is entangled with the key. The same correlations, measured across 1 200 km by the **Micius** satellite, are the record distance for entanglement distribution — Part D at continental scale.

6 Part E: Guided Activities

Simulator

All activities below run on the Optical Bench of the lab's interactive page ([lab7_optics.html](#)); the preset to use is named in each activity. All runs are seeded — the same seed always reproduces the same counts. Activity 6 uses the companion Qiskit notebook:

https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab7/lab7_optics.ipynb

6.1 Activity 1: Indivisibility Statistics (Preset A)

Student Activity

Fire 100, then 1 000, then 10 000 photons (same seed, resetting in between).

1. How does the deviation of the detector-B fraction from $1/2$ shrink as n grows? Compare with the statistical scale $\sqrt{1/(4n)}$.
2. Confirm the coincidence counter reads *exactly* 0 in every run — not just small. Why is this exact even though the simulation is random? (Equation (3).)
3. What would a classical wave-pulse model predict for the coincidence counter?

6.2 Activity 2: Mapping the Fringe (Preset B1)

Student Activity

Use the φ slider and the sweep button.

1. Verify the three anchor points: $\varphi = 0^\circ \rightarrow 100/0$, $\varphi = 60^\circ \rightarrow 75/25$, $\varphi = 180^\circ \rightarrow 0/100$ (Table 1).
2. Run the φ sweep and check the measured dots trace $\cos^2(\varphi/2)$. Where on the curve does a small phase change produce the largest change in counts? (This is where an interferometric sensor operates.)
3. Explain in one sentence why a photon that “chose one arm at BS1” cannot produce the $\varphi = 0$ result.

6.3 Activity 3: Buying Back the Fringes (Preset B2)

Student Activity

The complementarity dial: visibility $\mathcal{V} = s$.

1. At $s = 0.5$, verify the fringe extremes $0.75 / 0.25$ (fire at $\varphi = 0^\circ$ and 180°) and read the visibility card $\rightarrow 0.5$.
2. Run the φ sweep at $s = 1$, $s = 0.5$, $s = 0$ and compare the three curves' amplitudes.
3. Nothing ever reads the marker in this simulation. State precisely what kills the interference — and connect it to why Eve in Lab 6 cannot listen without leaving a trace.

6.4 Activity 4: Three Polarizers and the Staircase (Preset C)

Student Activity

1. Reproduce the pair of experiments: crossed $0^\circ/90^\circ \rightarrow$ exactly 0; insert $45^\circ \rightarrow 12.5\%$. Where do the absorbed photons go in each configuration? (Watch the per-polarizer absorption counts.)
2. With the $|H\rangle$ source and a single polarizer at 60° , verify Malus: 25% pass.
3. Set the staircase $0^\circ/30^\circ/60^\circ/90^\circ$ and check $\frac{1}{2}(3/4)^3 = 27/128 \approx 21.1\%$. Explain why *more* absorbing filters transmit *more* light, using the phrase “post-measurement state”.

6.5 Activity 5: The Factor of 2 (Preset D — Optional)

Student Activity

1. Set $\alpha = \beta = 20^\circ$: verify each side's own counter is $\approx 50/50$ while agreement is 100%. Change the common angle; does anything change?
2. Set $\alpha = 0^\circ$, $\beta = 22.5^\circ$ and verify $E \approx 0.707$ against the overlay. At $\Delta = 45^\circ$, why is $E = 0$? (Lab 6 vocabulary: what are the two analyzers to each other?)
3. Run the Δ sweep. The dashed curve is the best classical-wave story. Identify the angle where the quantum/classical gap is largest, and say what an experimenter would conclude from data landing on the solid curve.

6.6 Activity 6: The Optics in Qiskit

The companion notebook rebuilds this guide's optics as quantum circuits: the beam splitter as a one-qubit gate on the dual-rail path qubit, the Mach-Zehnder as a three-gate circuit swept over φ against $\cos^2(\varphi/2)$, which-path marking as a controlled-NOT onto an ancilla with the visibility sweep $\mathcal{V} = s$, the polarization experiments with Procedure 1 and Procedure 2 side by side, and (optional) the entangled-pair correlation curve against $\cos 2\Delta$. Its core circuit is three gates long:

```
1 def mach_zehnder(phi: float) -> QuantumCircuit:
2     """Mach-Zehnder on the dual-rail path qubit: BS - phase - BS."""
3     qc = QuantumCircuit(1)
4     qc.rx(-np.pi / 2, 0) # beam splitter B (the i-on-reflection 50/50)
```

```

5 qc.p(phi, 0)           # phase shifter Phi(phi) on arm B
6 qc.rx(-np.pi / 2, 0)  # beam splitter B
7 qc.measure_all()
8 return qc

```

D1 collects the photons observed in path B (outcome 1), so the counts of outcome 1 trace $P(D1) = \cos^2(\varphi/2)$ across the sweep. All notebook experiments are seeded and reproducible; expected statistics and tolerances are stated inline. A compact optional section runs the interferometer on real IBM hardware.

Important

Qiskit bit-ordering note. Qiskit orders multi-qubit bitstrings little-endian (rightmost bit = qubit q_0), while this lab series writes states big-endian (qubit 0 leftmost). Unlike Lab 6, where every register held a single qubit, this lab genuinely needs the conversion: two-qubit registers appear in the which-path model (the $|\text{path marker}\rangle$ states) and in Part D (the photon pairs), so *every* displayed bitstring must be converted to the leftmost-first ordering. The notebook shows exactly how, and applies it to every histogram.

6.7 Activity 7: Quick Check

Quick Check

Question 1: A single photon in state $(|A\rangle + i|B\rangle)/\sqrt{2}$ meets two detectors, one per path. What is the probability both click?

- (a) $1/4$ (b) exactly 0 (c) small but nonzero

Question 2: A Mach–Zehnder interferometer at $\varphi = 0$ is built from two 50/50 beam splitters. What fraction of photons reaches D1?

- (a) 50% — each splitter is a fair coin.
 (b) 100% — the two path amplitudes interfere.
 (c) 75%.

Question 3: A photon passes a polarizer at 45° (Procedure 1) and, in a second experiment, a half-waveplate at 22.5° + polarizing beam splitter (Procedure 2), in both cases with the “+” outcome. The outcome *probabilities* were identical. The exiting photon is...

- (a) the same state in both cases — probabilities fix the state.
 (b) $|+\rangle$ after Procedure 1, $|H\rangle$ after Procedure 2 — overlap only $1/2$.
 (c) unpolarized in both cases.

7 Resources

- Grangier, P., Roger, G. & Aspect, A., “Experimental evidence for a photon anticorrelation effect on a beam splitter: a new light on single-photon interferences,” *Europhys. Lett.* **1**, 173 (1986). — Part A’s anti-correlation and Part B’s single-photon interference in one experiment.

- Zeilinger, A., “General properties of lossless beam splitters in interferometry,” *Am. J. Phys.* **49**, 882 (1981). — Beam-splitter unitarity and the phase convention of Part A.
- Loudon, R., *The Quantum Theory of Light*, 3rd ed., Oxford University Press, 2000, §§5–6. — Beam splitters, single-photon states, and the $g^{(2)}$ correlation cited in Part A.
- Englert, B.-G., “Fringe visibility and which-way information: an inequality,” *Phys. Rev. Lett.* **77**, 2154 (1996). — The duality relation $\mathcal{V}^2 + \mathcal{D}^2 \leq 1$ behind Part B’s remark.
- Kwiat, P. G. et al., “New high-intensity source of polarization-entangled photon pairs,” *Phys. Rev. Lett.* **75**, 4337 (1995). — SPDC pair sources, Part D’s physical platform.
- Bell, J. S., “On the Einstein Podolsky Rosen paradox,” *Physics* **1**, 195 (1964); Clauser, J. F., Horne, M. A., Shimony, A. & Holt, R. A., *Phys. Rev. Lett.* **23**, 880 (1969). — Cited in Part D to back the “no classical story” claim; derivations beyond this lab’s scope.
- Nielsen, M. A. & Chuang, I. L., *Quantum Computation and Quantum Information*, Cambridge University Press, 2010, §7.4. — Optical photon quantum computation (the dual-rail encoding of Part A).
- Kok, P. et al., “Linear optical quantum computing with photonic qubits,” *Rev. Mod. Phys.* **79**, 135 (2007). — The LOQC/KLM context for Part A’s industry anchor.
- Hong, C. K., Ou, Z. Y. & Mandel, L., *Phys. Rev. Lett.* **59**, 2044 (1987). — Hong–Ou–Mandel interference: further reading, beyond this lab’s scope.
- Lab 5: *Quantum Measurements* and Lab 6: *Quantum Key Distribution (BB84)* (this series). — Born rule, Procedures 1 and 2, mutual unbiasedness, and the measurement–disturbance–security bridge.
- Companion Jupyter notebook: `quantum-labs-2026/lab7/lab7_optics.ipynb` (Colab link in Part E).